

MatrixLM: A Julia package to obtain closed-form least squares estimates for matrix linear models.

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DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

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Editor: Mehmet Hakan Satman

Reviewers:

- @palday
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Submitted: 05 February 2026

Published: unpublished

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Summary

MatrixLM is an open-source Julia package for fitting matrix linear models, which extend classical linear regression to a bilinear framework for matrix-valued responses. It is designed for analyzing high-throughput assays in which both rows and columns of the data matrix have associated covariates, such as in metabolomics, proteomics, or chemical genetic screens.

In a matrix linear model, the entries of a response matrix are modeled as a joint function of sample-level covariates (e.g., treatment group, demographic factors) and feature-level covariates (e.g., molecular or anatomical annotations, biological groups, pathways). MatrixLM implements efficient estimation and inference for this class of models using efficient matrix operations whenever possible allowing users to fit large numbers of models while retaining an explicit linear model interpretation. The inputs include a response matrix and two design matrices encoding the row and column covariates, and the main outputs include estimated coefficients, standard errors, and test statistics for user-specified contrasts.

Compared with workflows built from many separate univariate models, MatrixLM provides a unified interface for specifying, fitting, and summarizing matrix linear models. This reduces code duplication, improves reproducibility, and makes it easier to express hypotheses that naturally involve both sample- and feature-level information (for example, testing for differential effects across feature groups or experimental conditions). By providing an efficient implementation in Julia, MatrixLM enables researchers to perform interpretable analyses of large structured matrix-valued data.

Statement of need

High-throughput studies in biology and medicine often produce matrix-shaped data where each row corresponds to a sample (e.g., a patient, mutant strain, or experimental unit) and each column represents a molecular measurement (e.g., metabolite, gene, or phenotype). In many applications, both the samples and the measured features have associated metadata that should be incorporated into the analysis. However, existing tools either ignore these annotations or handle them in a fragmented, two-step fashion.

Standard approaches often involve fitting a separate model to each feature (e.g., using t-tests or linear models) and then performing a second-stage enrichment or grouping analysis. This approach is limited in two key ways: (1) it does not handle overlapping or quantitative annotations well, and (2) it fails to exploit shared structure among features or among samples. Dimension-reduction and machine learning methods address some of these issues, but often sacrifice interpretability and do not provide familiar statistical outputs like effect sizes or

41 confidence intervals.

42 MatrixLM addresses this gap by implementing matrix linear models (MLMs), a class of
43 bilinear models that allow researchers to directly model associations between sample-level and
44 feature-level characteristics. MLMs naturally accommodate both categorical and continuous
45 annotations, support hypothesis testing, and enable users to assess the effect of covariates
46 while adjusting for confounding structure. Compared to standard univariate workflows, MLMs
47 offer better interpretability and power, especially when annotations overlap or when feature
48 relationships are complex, as demonstrated in both chemical genetic screens (Liang et al.,
49 2019) and metabolomics applications (Farage et al., 2025).

50 Despite their utility, matrix linear models have not been widely available in reusable, general-
51 purpose software. MatrixLM provides a fast, open-source Julia implementation with a user-
52 friendly formula interface, making it easier for applied researchers to fit, interpret, and extend
53 these models in large-scale studies.

54 State of the field

55 The best-known software ecosystem for high-throughput biological data analysis is Bioconductor
56 (Huber et al., 2015). We are not aware of any package that provides the ability for large-scale
57 bilinear models in this ecosystem. The LIMMA family of packages (Ritchie et al., 2015),
58 fits separate linear models for each feature and borrows information across features using
59 empirical Bayes methods. While they use sample or individual-level information, they do not
60 use feature annotations in their modeling, which is central to MLMs. In the plant breeding
61 literature, bilinear models are often used for assessing gene-environment interactions. The best
62 known packages are metan (Olivoto & Lúcio, 2020) and Bilinear (Santantonio, 2022), but
63 neither is suited for high-throughput phenotypes, although they provide other more specialized
64 features such as missing data imputation using the EM algorithm. In principle, general-purpose
65 packages such as GLM.jl (Bates & others, 2024) and MixedModels.jl (Alday & Bates, 2026)
66 could be used to fit bilinear models using the Kronecker product formulation of MLMs. In
67 practice, this is impractical because the Kronecker product approach has memory requirements
68 beyond standard workstation hardware.

69 MatrixLM addresses these gaps by providing a dedicated Julia implementation for high-
70 throughput biological data that avoids explicit Kronecker-product formulations. It also offers a
71 clear interface for specifying row and column covariates, efficient estimation, and interpretable
72 statistical summaries.

73 Software design

74 MatrixLM is designed around two main goals: preserving interpretability and making matrix
75 linear models practical and fast for use on high-throughput data.

76 In addition to leveraging the speed of the Julia programming language, a central design choice
77 in MatrixLM is to exploit the algebraic structure of matrix linear models to compute least-
78 squares estimates directly, rather than relying on a generic regression engine built on an explicit
79 Kronecker-product design matrix. This reduces memory load and improves computational speed.
80 The implementation leverages Julia's LinearAlgebra.jl and its integration with OpenBLAS.
81 This allows MatrixLM to offload heavy matrix multiplications to highly optimized routines.

82 In addition to estimation, the package provides standard errors, t-statistics, and p-values for
83 the estimated coefficients. For high-throughput data where standard assumptions such as
84 normality may not hold, or for small sample sizes, MatrixLM also supports permutation testing,
85 offering a flexible alternative to traditional parametric tests.

86 The package additionally emphasizes usability through a model formula specification for

87 constructing the row and column design matrices. This lowers the barrier for researchers
88 who are already familiar with standard regression workflow syntax, while still allowing flexible
89 encodings of both categorical and continuous annotations.

90 Together, these design choices favor clarity, extensibility, and domain-oriented model
91 specification.

92 Mathematical framework

93 Matrix linear models extend ordinary linear regression to situations where the outcome is a
94 whole matrix rather than a single response vector. We arrange the data as follows:

- 95 ■ Y is an $n \times m$ matrix of high-throughput measurements (rows = samples, columns =
96 features).
- 97 ■ X is an $n \times p$ matrix of sample-level covariates (e.g., treatment group, sex, clinical
98 variables).
- 99 ■ Z is an $m \times q$ matrix of feature-level covariates (e.g., metabolite class, pathway, or other
100 annotations).
- 101 ■ B is a $p \times q$ matrix of regression coefficients linking the sample and feature covariates.

102 The matrix linear model assumes

$$Y = XBZ^T + E,$$

103 where E is an $n \times m$ matrix of residuals. In element-wise form, each entry y_{ij} is written as

$$y_{ij} = \sum_{k=1}^p \sum_{\ell=1}^q x_{ik} z_{j\ell} b_{k\ell} + e_{ij}.$$

104 As the equation above shows, the elements of B may be interpreted as interactions between
105 the columns of X and the columns of Z . The residuals are assumed to have zero mean, with
106 a covariance Σ across the features and independent across samples,

$$V(\text{vec}(E)) = \Sigma \otimes I.$$

107 The current implementation of MatrixLM requires complete response and predictor matrices.
108 Missing data should therefore be addressed before model fitting using an appropriate method
109 selected according to the assumed missingness mechanism and the structure of the data.

110 We treat X and Z as known and estimate B by least squares, choosing $\hat{B}$ to minimize the
111 Frobenius norm of the residuals,

$$\hat{B} = \arg \min_B \|Y - XBZ^T\|_F^2.$$

112 This optimization problem has a closed-form solution:

$$\hat{B} = (X^T X)^{-1} X^T Y Z (Z^T Z)^{-1},$$

113 when $X^T X$ and $Z^T Z$ are invertible. From $\hat{B}$, the software can construct fitted values
114 $\hat{Y} = X\hat{B}Z^T$ and standard errors and test statistics for entries of $\hat{B}$ or for user-specified linear
115 contrasts, directly analogous to classical linear models. The covariance matrix Σ is estimated
116 from the empirical covariance of the residuals. Since the number of features can be comparable
117 or greater than the number of samples, the covariance estimate can be singular. The package
118 provides the user the option of using a shrinkage estimator for the covariance matrix which is
119 non-singular.

120 When there is only a single feature (one column of Y) and no feature-level design matrix Z ,
121 this framework reduces to ordinary linear regression with design matrix X . The MatrixLM
122 package therefore generalizes familiar linear modeling ideas to matrix-valued outcomes with
123 structured annotations on both rows and columns.

124 Figures

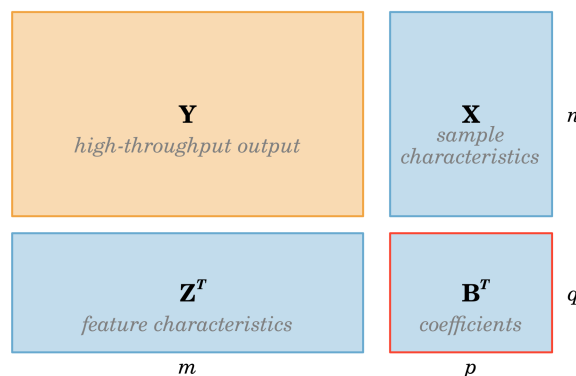


Figure 1: A visualization of the response ($Y : n \times m$), sample covariates ($X : n \times p$), feature covariates ($Z : m \times q$), and coefficients ($B : p \times q$) matrices for a matrix linear model. The dimensions in the model correspond to n samples/individuals, m features/measurements, p sample covariates, and q feature covariates; the matrix B is to be estimated.

125 A visualization of the response ($Y : n \times m$), sample covariates ($X : n \times p$), feature covariates
126 ($Z : m \times q$), and coefficients ($B : p \times q$) matrices for a matrix linear model. The dimensions in the model
127 correspond to n samples/individuals, m features/measurements, p sample covariates, and q feature covariates;
128 the matrix B is to be estimated.

129 Research impact statement

130 MatrixLM provides a practical implementation of matrix linear models for encoding relationships
131 and groupings high-throughput matrix-shaped data with annotations on both samples and
132 features. By combining a flexible formula interface with fast, closed-form least-squares
133 estimation, the package makes it straightforward for applied researchers to encode biological or
134 experimental structure directly into their models and apply matrix linear models in reproducible
135 workflows.

136 Conclusion and future directions

137 In ongoing work, we are extending this framework to penalized matrix linear models for high-
138 dimensional settings. In particular, we are developing a companion Julia package, MatrixLMnet,
139 which implements elastic-net and related penalties on the coefficient matrix to support variable
140 selection and regularization in matrix linear models.

141 AI usage disclosure

142 Grammarly was used during the drafting of this manuscript to assist with linguistic polishing.
143 The authors used GitHub Copilot to assist in coding testing functions. All Copilot-suggested
144 code was reviewed, tested, and validated by the human authors to ensure correctness. The
145 authors take full responsibility for the content of this manuscript.

Acknowledgements

This work started when JWL was a summer intern at UCSF, and continued when she was a scientific programmer at the University of Tennessee Health Science Center (UTHSC). We thank both UCSF and UTHSC for funding, and a supportive environment. This work was partly supported by National Institutes of Health grants GM-070683 (SS), GM-078338 (SS), GM-123489 (SS, GF), ES-022841 (SS), and DA-044223 (SS).

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